

SALONY VERMA

Chandigarh, India (Open to Relocate to Germany)

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Summary

Genomics and clinical microbiology scientist with 10+ years of experience across AMR genomics, molecular diagnostics, and regulated laboratory environments. Proven expertise in whole genome sequencing, targeted NGS, PK/PD modeling, and audit-ready scientific documentation within BSL-2 and BSL-3 labs. Experienced in SOP development, quality systems and regulatory-aligned scientific reporting for clinical and academic research organizations.

Core Skills

- **Genomics & Molecular:** WGS, Targeted NGS, qRT-PCR, DNA/RNA extraction
- **Data & Analysis:** GraphPad Prism, SPSS, PK/PD modeling, statistical reporting
- **Clinical Microbiology:** AST/DST, phenotypic– genotypic AMR correlation
- **Quality & Compliance:** SOPs, QC records, audit readiness, GLP documentation
- **Platforms & Instruments:** Illumina MiSeq, ONT, Qubit, Bioanalyzer, GeneXpert
- **Scientific Writing:** Technical reports, manuscripts, regulatory documentation

Experience

Panjab University

October 2024 – October 2025

Guest Faculty (Assistant Professor)

Chandigarh, India

- Conducted postgraduate teaching in microbial biotechnology and applied genomics.
- Developed SOPs, lab manuals, and assessment frameworks to standardize QC and documentation.
- Accelerated thesis completion by mentoring 10+ students and guiding 6 MSc projects per cycle.
- Improved academic compliance through structured reporting aligned with institutional guidelines.

PGIMER

March 2024 – July 2024

Postdoctoral Research Associate, Medical Microbiology

Chandigarh, India

- Supported regulated mycobacteriology workflows for diagnostic and research decision-making.
- Executed Deeplex Myc-TB sequencing for clinical isolates, including DNA extraction and library QC reporting.
- Reduced repeat sequencing by resolving contamination flags and improving run readiness metrics.
- Delivered audit-ready technical summaries for internal clinical and research stakeholders.

MM College of Medical Sciences and Research

June 2023 – October 2023

Medical Microbiology Tutor

Ambala, India

- Taught Medical Microbiology to MBBS students, integrating core theory with routine diagnostic laboratory practices.
- Contributed to the establishment of bacteriology, immunology, serology, virology, parasitology, and mycology laboratory sections.
- Participated in routine diagnostic microbiology testing and verification of laboratory reports under departmental protocols.
- Maintained quality control records and laboratory documentation during inspections by NABH, CGHS, and ECHS.

School of Life Sciences, Desh Bhagat University

February 2023 – June 2023

Assistant Professor (Microbiology)

Gobindgarh, India

- Delivered applied microbiology instruction emphasizing diagnostic laboratory practices.
- Guided PhD scholar on experimental design, literature synthesis, and execution planning.
- Strengthened academic delivery through assessment design and practical evaluations

PGIMER

June 2021 – June 2022

Senior Research Fellow, AMR Genomics

Chandigarh, India

- Investigated carbapenem and colistin resistance in *Italicise Escherichia coli* using phenotypic testing and whole-genome sequencing approaches.
- Standardized DNA and library QC SOPs using Qubit, Bioanalyzer, and gel electrophoresis.
- Sequenced 100+ isolates on Illumina and ONT with reproducible, audit-ready workflows

PGIMER

March 2020 – October 2021

Laboratory Instructor, COVID-19 Diagnostics

Chandigarh, India

- Scaled national SARS-CoV-2 diagnostics using qRT-PCR and GeneXpert platforms.
- Trained 200+ personnel on SOP adherence, QC controls, and documentation compliance.
- Reduced operator variability via competency assessments and standardized work instructions.

PGIMER

January 2016 – June 2021

PhD Researcher, Medical Microbiology

Chandigarh, India

- Conducted research on PK/PD of minocycline in bronchopulmonary infection.
- Quantified drug levels via SPE and HPLC with validated run controls.
- Modeled exposure-response in Phoenix WinNonlin and NONMEM to support data interpretation.
- Prepared scientific abstracts and research reports and presented findings at international conferences

Panjab University

September 2014 – December 2015

Junior Research Fellow

Chandigarh, India

- Performed diagnostic assays for pathogen detection studies.
- Maintained GLP- compliant documentation, equipment calibration records, and laboratory inventory.

Education

Postgraduate Institute of Medical Education and Research (PGIMER)

2021

PhD, Medical Microbiology

Chandigarh, India

Harvard Medical School

2020

HMX Fundamentals of Pharmacology

Remote

Panjab University

2013

MSc, Microbiology

Chandigarh, India

Panjab University

2011

BSc, Microbiology

Chandigarh, India

Certifications & Awards

- Bill & Melinda Gates Foundation Travel Award, ASM Microbe (2021)
- ICMR Senior Research Fellowship (2019) and Junior Research Fellowship (2016)
- UGC-BSR Fellowship (2014)
- GATE (2013), 94 Percentile

Languages

- English: Full Professional Proficiency (C2)
- German: Basic Proficiency (A1)